R Lab – First 15 Programs

Programs with comments removed.

1. Create variables of different types and identify their types.

```
name <- "Alice"
age <- 25
height <- 5.6
is_student <- TRUE

name
age
height
is_student

class(name)
class(age)
class(height)
class(is_student)
```

2. Define 2 numbers. Perform all arithmetic operations in R.

```
a <- 15
b <- 4

addition <- a + b
cat("Addition: ", a, "+", b, "=", addition, "\n")

subtraction <- a - b
cat("Subtraction: ", a, "-", b, "=", subtraction, "\n")

multiplication <- a * b
cat("Multiplication: ", a, "*", b, "=", multiplication, "\n")

division <- a / b
cat("Division: ", a, "/", b, "=", division, "\n")

int_division <- a %/% b
cat("Integer Division: ", a, "%/%", b, "=", int_division, "\n")

modulus <- a %% b
cat("Modulus: ", a, "%%", b, "=", modulus, "\n")

power <- a ^ b
cat("Exponentiation: ", a, "^", b, "=", power, "\n")
```

3. Print the multiplication table of 5 using for loop.

```
num <- 5

cat("Multiplication Table of ", num, "\n")

for (i in 1:10) {
    result <- num * i
    cat(num, "x", i, "=", result, "\n")
}
```

4. Find the largest number among 3 numbers.

```
a <- 10
b <- 25
c <- 15

big <- a

if (b > big) {
    big <- b
}

if (c > big) {
    big <- c
}
```

```
cat("The largest number is:", big)
```

5. Count the even and odd numbers in a vector.

```
numbers <- c(12, 5, 7, 8, 14, 3, 9, 10)

even_count <- 0
odd_count <- 0

for (num in numbers) {
    if (num %% 2 == 0) {
        even_count <- even_count + 1
    } else {
        odd_count <- odd_count + 1
    }
}

cat("Even numbers:", even_count, "\n")
cat("Odd numbers:", odd_count, "\n")
```

6. Create a numeric vector with 10 elements. Perform the following:

```
a) Access the first element
b) Access the second element to sixth element
c) Modify third element to 88
d) Add 5 to each element
e) List the elements greater than 50
f) Find the average of the values
g) Find the maximum value
h) Find the minimum value
i) Find the median
j) Find the range
k) Find the standard deviation

v <- c(12, 45, 67, 23, 89, 34, 56, 78, 10, 90)

cat("a) First element:", v[1], "\n")
cat("b) Second to sixth elements:", v[2:6], "\n")

v[3] <- 88
cat("c) Vector after modifying third element:", v, "\n")

v_plus_5 <- v + 5
cat("d) Vector after adding 5 to each element:", v_plus_5, "\n")

cat("e) Elements greater than 50:", v[v > 50], "\n")
cat("f) Average (mean):", mean(v), "\n")
cat("g) Maximum value:", max(v), "\n")
cat("h) Minimum value:", min(v), "\n")
cat("i) Median value:", median(v), "\n")
cat("j) Range (min and max):", range(v), "\n")
cat("k) Standard deviation:", sd(v), "\n")
```

7. Create a list named student with name, age and scores of 3 subjects. Access the list elements. Find the average score.

```
student <- list(name = "John", age = 22, scores = c(88, 90, 85))

student$name
student$age
student$scores

mean(student$scores)
```

8. Create a dataframe of students with name, age and marks of 5 students. Perform the following:

```
a) View the entire dataframe
b) Access only the column Name
c) Access the second row
d) Access the mark of third student
e) Add a new column Grade based on marks
f) List students who secured more than 80 marks
g) Find the average marks

students <- data.frame(
    Name = c("Alice", "Bob", "Carol", "David", "Eva"),
    Age = c(20, 21, 19, 22, 20),
    Marks = c(85, 76, 90, 65, 88)
)

cat("a) Entire Data Frame:\n")
students

cat("\nb) Names of Students:\n")
students$Name

cat("\nc) Second Row:\n")
students[2, ]

cat("\nd) Marks of Third Student:\n")
students$Marks[3]

students$Grade <- ifelse(
```

```
    students$Marks >= 85, "A",
    ifelse(students$Marks >= 70, "B", "C")
)

cat("\ne) Data Frame with Grade Column:\n")
students

cat("\nf) Students with Marks > 80:\n")
subset(students, Marks > 80)

cat("\ng) Average Marks:\n")
mean(students$Marks)
```

9. Write an R script to concatenate 2 strings.

```
first_name <- "John"
last_name <- "Doe"

full_name <- paste(first_name, last_name)
cat("Full Name (with space):", full_name, "\n")

username <- paste0(first_name, last_name)
cat("Username (no space):", username, "\n")
```

10. Write an R script to extract a substring from a string.

```
text <- "Programming"

sub_part <- substr(text, 4, 7)

cat("Substring:", sub_part, "\n")
```

11. Write an R script to replace a substring from a string.

```
sentence <- "R is fast. R is flexible."

cat("Using sub():", sub("R", "Python", sentence), "\n")

cat("Using gsub():", gsub("R", "Python", sentence), "\n")
```

12. Write an R script to find the number of characters in a string. Also find the uppercase and lowercase of the string.

```
word <- "DataScience"

upper <- toupper(word)
lower <- tolower(word)

cat("Uppercase:", upper, "\n")
cat("Lowercase:", lower, "\n")
cat("Length of string:", nchar(word), "\n")
```

13. Consider a CSV file named marks2.csv. Read the CSV file and perform the following:

```
a) Display the first few rows.
b) Display the structure of the dataframe.
c) Perform the summary statistics.
d) Display the column names.
e) Display the number of rows and columns.
f) Access a specific column.
g) Display the average value of a column.
h) Display the median value of a column.
i) Display the standard deviation of a column.
j) Display the range of a column.

data <- read.csv("marks2.csv")

head(data)
str(data)
summary(data)
names(data)
nrow(data)
ncol(data)

data$Biology
mean(data$Biology, na.rm = TRUE)
median(data$Maths, na.rm = TRUE)
sd(data$Physics, na.rm = TRUE)
range(data$Physics, na.rm = TRUE)
```

14. Consider a CSV file called Sales_by_Region.csv with the following data. Draw a Pie chart using the given data.

```
Region,Sales
North,15000
South,10000
East,8000
West,12000
Central,5000

sales <- read.csv("Sales_by_Region.csv")

pie(
    sales$Sales,
    labels = sales$Region,
    main = "Sales Distribution by Region",
    col = rainbow(length(sales$Sales))
)

pct <- round(sales$Sales / sum(sales$Sales) * 100)
labels <- paste(sales$Region, "-", pct, "%")

pie(
    sales$Sales,
    labels = labels,
    main = "Sales Distribution by Region",
    col = rainbow(length(sales$Sales))
)
```

15. Analyze sales data for a grocery store. The store sells five types of fruits and we want to visualize their total sales. Hint: Use bar chart.

```
fruits <- c("Apple", "Banana", "Mango", "Orange", "Grapes")
sales <- c(150, 200, 120, 180, 90)

bar_positions <- barplot(
    sales,
    names.arg = fruits,
    col = "skyblue",
    main = "Fruit Sales in a Grocery Store",
    xlab = "Fruit Type",
    ylab = "Units Sold",
    border = "blue",
    ylim = c(0, max(sales) + 50)
)
```

```
text(
    x = bar_positions,
    y = sales + 5,
    labels = sales,
    col = "red"
)
```

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Programs without comments

16. CSV File

```
students <- data.frame(
  ID = c(1, 2, 3),
  Name = c("John", "Alice", "Bob"),
  Marks = c(85, 90, 78)
)
write.csv(students, "output.csv", row.names = FALSE)
df <- read.csv("output.csv", header = TRUE)
print(df)
```

17. Text File

```
students <- data.frame(
  ID = c(1, 2, 3),
  Name = c("John", "Alice", "Bob"),
  Marks = c(85, 90, 78)
)
write.table(students, "students.txt")
students <- read.table("students.txt", header = TRUE)
print(students)
```

18. Excel File

```
install.packages("readxl")
install.packages("openxlsx")

library(readxl)
library(openxlsx)

students <- data.frame(
  ID = c(1, 2, 3),
  Name = c("John", "Alice", "Bob"),
  Marks = c(85, 90, 78)
)

write.xlsx(students, "people.xlsx")
data <- as.data.frame(read_excel("people.xlsx"))
print(data)
```

19. Scatter Plot

```
x <- c(10, 20, 30, 40, 50)
y <- c(15, 25, 28, 43, 55)

plot(x, y,
     main = "Scatter Plot Example",
     xlab = "X Values",
     ylab = "Y Values",
     col = "blue",
     pch = 16,
     cex = 1.2)
```

20(a). Box and Whiskers Plot

```
scores <- c(65, 70, 75, 80, 85, 90, 95, 100, 105, 110)
boxplot(scores,
```

```
        main = "Score Distribution",
        ylab = "Scores",
        col = "skyblue",
        border = "red",
        notch = TRUE)
```

20(b). Multiple Boxplots

```
math <- c(60, 65, 70, 75, 80)
science <- c(68, 72, 74, 76, 85)
english <- c(55, 60, 65, 70, 75)

scores <- list(
  Math = math,
  Science = science,
  English = english
)

boxplot(scores,
        main = "Scores by Subject",
        col = c("red", "green", "blue"),
        ylab = "Marks")
```

21(a). Simple Bar Chart

```
values <- c(25, 40, 35, 50)
categories <- c("A", "B", "C", "D")
colors <- c("red", "blue", "green", "orange")

barplot(values,
        names.arg = categories,
        col = colors,
        main = "Category-wise Values",
        xlab = "Categories",
        ylab = "Values",
        border = "black")
```

21(b). Grouped Bar Chart

```
quarters <- c("Q1", "Q2", "Q3", "Q4")
fruits <- c(100, 120, 150, 50)
vegetables <- c(80, 95, 110, 175)
grocery <- c(120, 135, 140, 75)

data_matrix <- rbind(fruits, vegetables, grocery)
colors <- c("red", "green", "blue")

barplot(data_matrix,
        beside = TRUE,
        names.arg = quarters,
        col = colors,
        main = "Sales by Quarter and Category",
        xlab = "Quarter",
        ylab = "Sales",
        border = "black")

legend("topleft",
       legend = rownames(data_matrix),
       fill = colors)
```

21(c). Stacked Bar Chart

```
quarters <- c("Q1", "Q2", "Q3", "Q4")
fruits <- c(100, 120, 150, 50)
```

```
vegetables <- c(80, 95, 110, 175)
grocery <- c(120, 135, 140, 75)

data_matrix <- rbind(fruits, vegetables, grocery)
colors <- c("red", "green", "blue")

barplot(data_matrix,
        beside = FALSE,
        names.arg = quarters,
        col = colors,
        main = "Sales by Quarter and Category",
        xlab = "Quarter",
        ylab = "Sales",
        border = "black")

legend("topleft",
       legend = rownames(data_matrix),
       fill = colors)
```

22. Histogram

```
scores <- c(
  45, 50, 55, 60, 62, 65, 67, 70, 72, 75,
  78, 80, 82, 85, 88, 90, 92, 95, 68, 74,
  58, 63, 77, 81, 69, 73, 84, 87, 91, 96
)

hist(scores,
     main = "Distribution of Student Exam Scores",
     xlab = "Marks",
     ylab = "Number of Students",
     col = "lightblue",
     border = "black")
```

23(a). Simple Linear Regression

```
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)

plot(y, x,
     col = "blue",
     main = "Height & Weight Regression",
     abline(lm(x ~ y)),
     cex = 1.3,
     pch = 16,
     xlab = "Weight in Kg",
     ylab = "Height in cm")
```

23(b). Linear Regression Model

```
relation <- lm(y ~ x)
print(relation)
```

23(c). Prediction

```
a <- data.frame(x = 170)
result <- predict(relation, a)
print(result)
```

24(a). Multiple Linear Regression

```
input <- mtcars[, c("mpg", "disp", "hp", "wt")]
print(head(input))
```

24(b). Multiple Regression Model

```
model <- lm(mpg ~ disp + hp + wt, data = input)
print(model)

cat("#### The Coefficient Values ####\n")

a <- coef(model)[1]
print(a)

Xdisp <- coef(model)[2]
Xhp <- coef(model)[3]
Xwt <- coef(model)[4]

print(Xdisp)
print(Xhp)
print(Xwt)
```

24(c). Multiple Regression Prediction

```
new_car <- data.frame(
  disp = 221,
  hp = 102,
  wt = 2.91
)

predicted_mpg <- predict(model, newdata = new_car)
predicted_mpg
```

25(a). Mean

```
x <- c(10, 2, 30, 4, 5, 6, 70, 8, 9, 10)
y <- c(1, 2, 3, 4, 5, NA)

mean(x)
mean(x, trim = 0.2)
mean(y, na.rm = TRUE)
```

25(b). Median

```
x <- c(10, 2, 30, 2, 5, 8, 70, 8, 9, 2)
y <- c(3, 2, 3, 4, 5, NA)

median(x)
median(y, na.rm = TRUE)
```

25(c). Range

```
x <- c(10, 2, 30, 4, 5, 6, 70, 8, 9, 10)
y <- c(1, 2, 3, 4, 5, NA)

range(x)
range(y, na.rm = FALSE)
```

25(d). Variance

```
x <- c(10, 2, 30, 2, 5, 8, 70, 8, 9, 2)
y <- c(3, 2, 3, 4, 5, NA)

var(x)
var(y, na.rm = TRUE)
```